EXPERIMENT NO 1

FIBONACCI RECCURSIVE

```
#include<iostream>
#include <chrono>
using namespace std;
using namespace std::chrono;
void printFibonacci(int n) {
static int n1 = 0, n2 = 1, n3;
if (n > 0) {
n3 = n1 + n2;
n1 = n2;
n2 = n3;
cout << n3 << " ";
printFibonacci(n - 1);
}
}
int main() {
int n;
cout << "Enter the number of elements: ";
cin >> n;
cout << "Fibonacci Series: ";
cout << "0 " << "1 ";
auto start_time = high_resolution_clock::now();
printFibonacci(n - 2); // n-2 because 2 numbers are already printed
auto end_time = high_resolution_clock::now();
auto duration = duration_cast<microseconds>(end_time - start_time);
cout << "\nElapsed Time: " << duration.count() << " microseconds" << endl;

// Additional space tracking
```

```
cout << "Estimated Space Used: " << sizeof(int) * 3 * (n - 2) << " bytes" <<
endl;
return 0;
}
```

OUTPUT

```
icoer@icoer-Vostro-3470:~$ g++ fibonacci_recur.cpp
icoer@icoer-Vostro-3470:~$ ./a.out
Enter the number of elements: 6
Fibonacci Series: 0 1 1 2 3 5
Elapsed Time: 5 microseconds
Estimated Space Used: 48 bytes
icoer@icoer-Vostro-3470:~$
```